

Note 15.7 Triple Integrals in Cylindrical and Spherical Coordinates

Integration in Cylindrical Coordinates

Definition

Cylindrical coordinates represent a point P in space by ordered triples (r, θ, z) where

- r and θ are polar coordinates for the vertical projection of P on the xy -plane
- z is the rectangular vertical coordinates.

Equations Relating Rectangular (x, y, z) and Cylindrical (r, θ, z) Coordinates

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$r^2 = x^2 + y^2 \quad \tan \theta = \frac{y}{x}$$

Integrals of Cylindrical Coordinates

$$\iiint_D f r \, dr \, d\theta \, dz = \iiint_D f \, dz \, r \, dr \, d\theta$$

Spherical Coordinates and Integration

Definition

Spherical coordinates represent a point P in space by ordered triples (ρ, ϕ, θ) where

- ρ is the distance from P to the origin.
- ϕ is the angle $\overrightarrow{OP}$ makes with the positive z -axis. ($0 \leq \phi \leq \pi$)
- θ is the angle from cylindrical coordinates. ($0 \leq \theta \leq 2\pi$)

Equations

$$r = \rho \sin \phi, \quad x = r \cos \theta = \rho \sin \phi \cos \theta$$

$$z = \rho \cos \phi, \quad y = r \sin \theta = \rho \sin \phi \sin \theta$$

$$\rho = \sqrt{x^2 + y^2 + z^2} = \sqrt{r^2 + z^2}$$

Integrals of Spherical Coordinates

$$\iiint_D f(\rho, \phi, \theta) \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

and

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$